

Anas Arfeen

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PROFESSIONAL SUMMARY

Computer Science undergraduate at VIT Chennai focused on artificial intelligence, reinforcement learning, and full-stack systems engineering. Builds production products with real users alongside research-driven AI agents and LLM tooling. Currently serving as AI/ML Co-Lead at the Microsoft Innovations Club.

EDUCATION

Vellore Institute of Technology (VIT)	Chennai, India
<i>B.Tech in Computer Science Engineering</i>	2025 – 2029
• AISSCE (CBSE) – 90%	2024–25
• AISSE (CBSE) – 94%	2022–23

TECHNICAL SKILLS

Languages: Python, C++, Java, C, C#, TypeScript/JavaScript, Go, R, MATLAB, HTML/CSS, Assembly (x86, basic)
AI/ML: PyTorch, Stable-Baselines3, PPO, CNNs, ResNet50, OpenCV, MediaPipe, RAG, LLM Integration (Gemini/Ollama)
Full-Stack: Next.js, FastAPI, PostgreSQL, Supabase, Docker
Embedded/Hardware: Arduino, ESP32, PlatformIO, Sensor/Servo Modules
Tools: Linux (Arch), Git, Neovim, VS Code
Core Areas: Reinforcement Learning, Deep Learning, Computer Vision, Systems Programming, Full-Stack Development, OOP, DSA

LEADERSHIP & EXPERIENCE

Microsoft Innovations Club (MIC)	VIT Chennai
<i>AI/ML Co-Lead</i>	2026 – Present
- Orchestrate AI/ML-focused workshops and technical initiatives for the student community. - Mentor students in modern development workflows, AI tooling, and project-building best practices. - Direct technical planning and automate tasks to streamline event execution and community engagement.	
<i>AI/ML Member</i>	2025 – 2026
Contributed to collaborative technical projects and participated in advanced AI learning sessions.	
Linux User Group (LUG)	VIT Chennai
<i>Technical Member</i>	2025 – 2026
Advocated for open-source ecosystems and optimized terminal-based developer productivity workflows across the club's Linux and dotfiles community.	
Hack Club	VIT Chennai
<i>AI/ML Member</i>	2025 – Present
Participates in student-led hackathons and rapid-prototyping events, building and shipping projects in collaborative coding sessions.	

PROJECTS

<u>Autonomous Warehouse Rover</u>	<i>Python, PyTorch, Stable-Baselines3</i>
- Engineered a PPO-based reinforcement-learning agent using policy-gradient optimization for autonomous navigation, trained in a custom Gymnasium environment with 15-ray lidar observations across 8 vectorized parallel environments. - Applied curriculum learning with staged promotion across four warehouse-density tiers, progressively increasing obstacle complexity as the agent's success rate improved.	
<u>poke-ai – LLM Pokémon Battle AI</u>	<i>Python, LLM Integration, Docker</i>
- Architected a hybrid decision engine fusing a low-latency heuristic move-scoring function with LLM-based strategic reasoning (Gemini/Ollama), balancing real-time constraints against deeper tactical planning. - Integrated the agent with a self-hosted Pokémon Showdown battle server via a custom web client, supporting live-swappable LLM backends and a containerized Docker deployment.	

Reverse Akinator

React, Express, RAG, LLM Integration

- Built a retrieval-augmented generation (RAG) pipeline over an offline knowledge base of 35+ footballer profiles, grounding the LLM's dynamically-generated yes/no questions in factual retrieval rather than a static decision tree.
- Designed a provider-agnostic LLM abstraction layer spanning 6 backends (Ollama, Gemini, OpenAI, Claude, Groq, Puter.js), orchestrated by a strict finite-state machine governing game flow.

Spell Masters – Gesture-Controlled Game

Python, OpenCV, MediaPipe

- Engineered a 72-dimensional fingertip/wrist gesture-embedding pipeline matched via cosine similarity for tilt-resistant, real-time gesture recognition.
- Decoupled the OpenCV/MediaPipe CV inference pipeline onto a background thread with a thread-safe event queue, keeping the Pygame render loop locked at 60FPS.

Relay – Club Sponsorship CRM

Next.js, TypeScript, Supabase

- Designed a multi-tenant CRM schema (PostgreSQL with Row-Level Security policies) isolating per-club data while sharing a global sponsor directory, replacing spreadsheet-based outreach with a Kanban pipeline.
- Integrated Gmail OAuth2 for in-app outreach-thread sync; owned the majority of system architecture and schema design on a 2-person team.

AmazeCC – Student Portal Dashboard

Next.js, TypeScript, PostgreSQL, FastAPI

- Built a ~130-route full-stack platform (Next.js/TypeScript frontend, FastAPI backend, PostgreSQL) unifying attendance, grades, hostel, and payments for VIT students – live in production at amazecc.com with real, active users.
- Shipped an installable PWA with offline caching and push notifications, alongside a separate ops-only admin dashboard for internal data management.

INTERESTS

Linux Customization, Terminal Ricing, Experimental Coding Projects, Coffee Culture, Music, Story-Driven Games, Online Tech Communities